

Ohomom.om29@gmail.com

0563651121

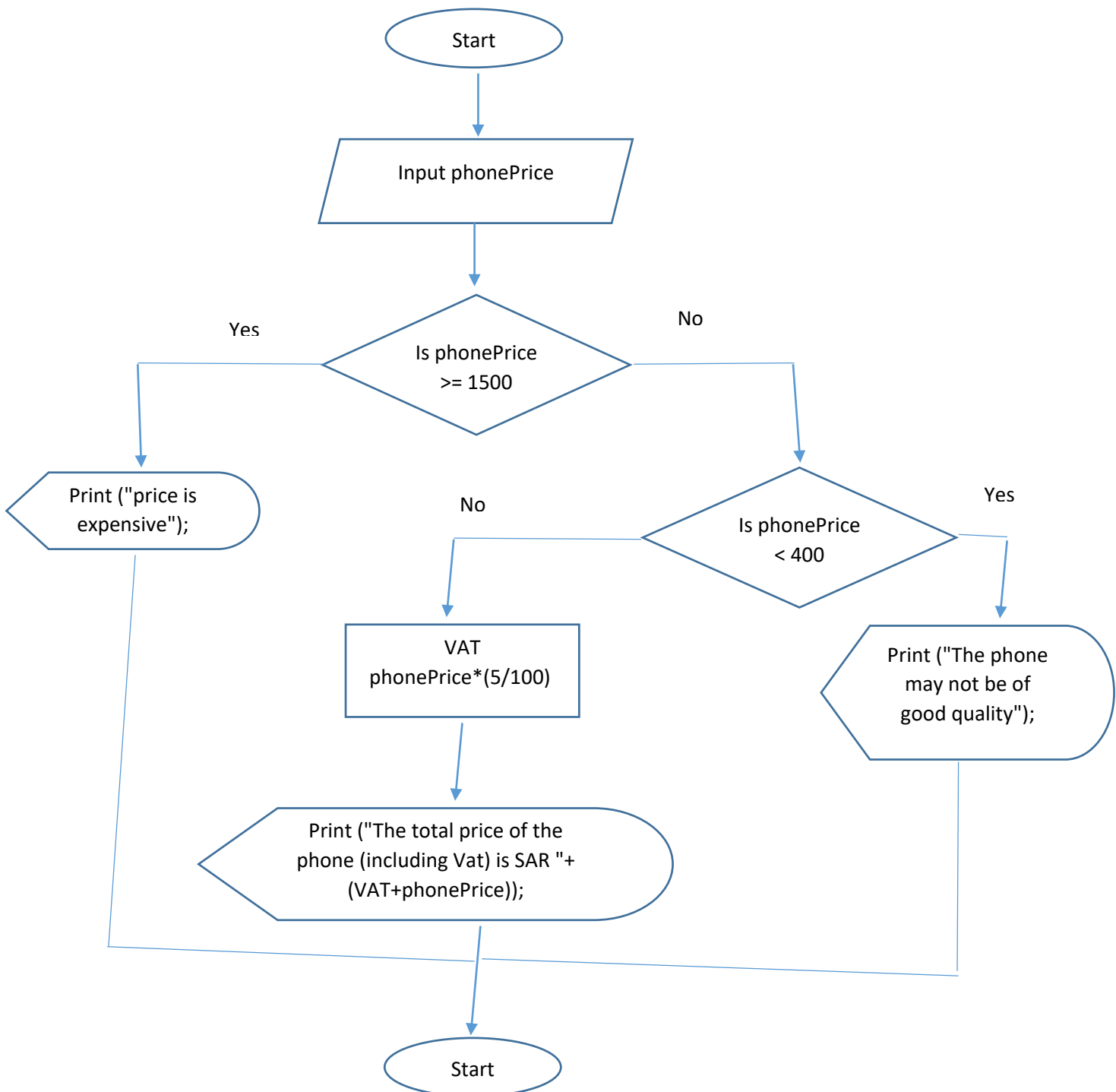
Ohoud Alhindi

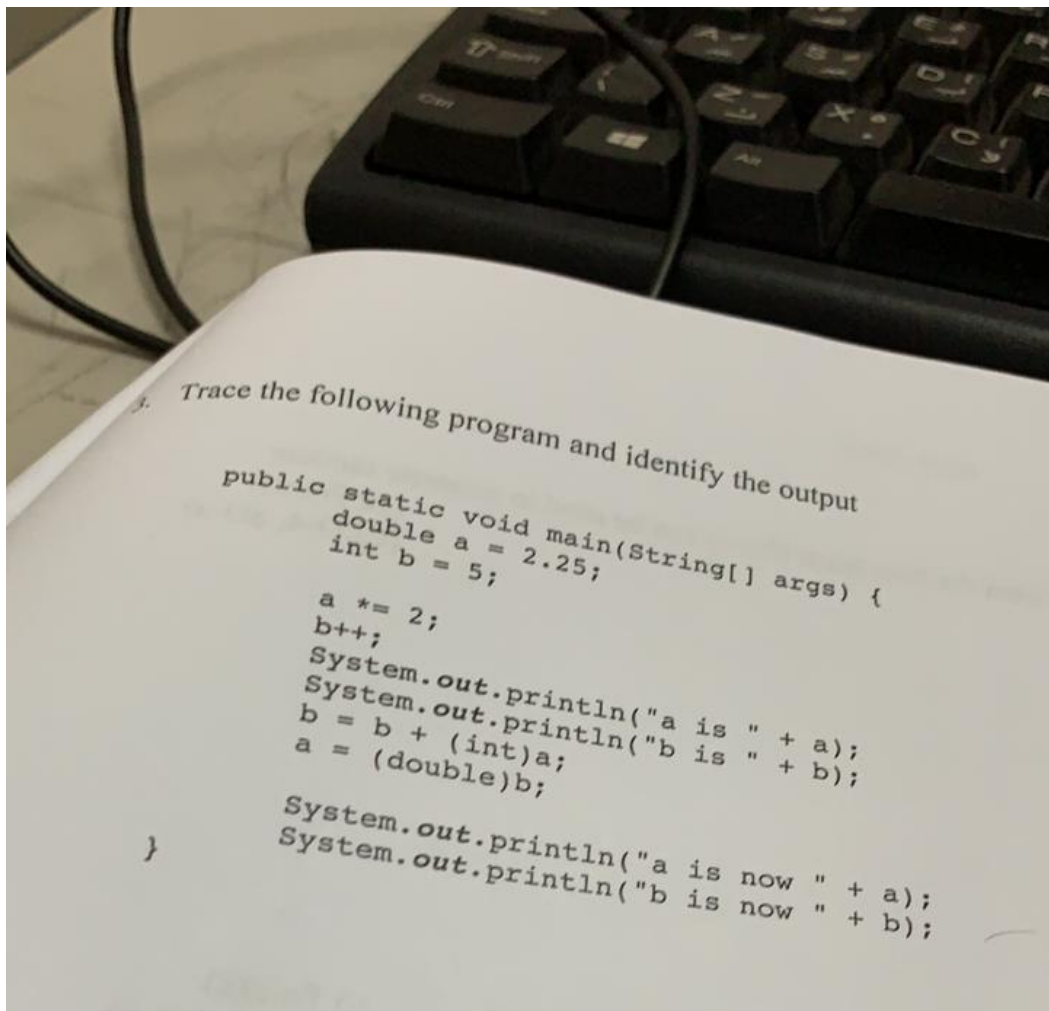
[التاريخ]

Part II: Short Answer Questions (40 points)

1. Describe the steps (in pseudo code or using a flowchart) needed to perform the following (10 points)
(CLO-1, SO-c)

Read the price of a phone from the user. If the price of the phone is greater than or equal to SAR1500, display the message "Price is expensive". If the price of the phone is below SAR400, display the message "The phone may not be of good quality". For all other prices calculate the VAT(5%) based on the price of the phone, add it to the price and display the total in a message of the form - "The total price of the phone (including VAT) is SAR _____"





3. Trace the following program and identify the output

```
public static void main(String[] args) {  
    double a = 2.25;  
    int b = 5;  
  
    a *= 2;  
    b++;  
    System.out.println("a is " + a);  
    System.out.println("b is " + b);  
    b = b + (int)a;  
    a = (double)b;  
  
    System.out.println("a is now " + a);  
    System.out.println("b is now " + b);  
}
```

a is 4.5

b is 6

a is now 10.0

b is now 10

2. Explain the behavior of the following code in simple English

```
public static void main(String[] args) {  
    Scanner input = new Scanner(System.in);  
    int number = input.nextInt();  
    switch(number)  
    {  
        case 211:  
        case 423:  
            System.out.println("Nice numbers");  
        case 41:  
        case 45:  
            System.out.println("Better numbers");  
            break;  
        case 62:  
        case 28:  
            System.out.println("Bad numbers");  
        default:  
            System.out.println("Smelly numbers");  
    }  
}
```

1. Create scanner object, to read value from user
2. declare int variable, its value will be entered by the user
3. switch statement for different conditions (number) values
4. it will execute in order, break means stop, if nothing true default will execute
5. if case without break executed all cases after it will execute till break is found or the end of switch.
6. Example if user enters 211
Case 211, 423, 41, 45 will execute
But if he enters 45
Only case 45 will execute

Part B: Multiple Choice (15 points, 3 points each)

(CLO-12, SO-a)

1. What will be the output of the code segment, after executing the following statements?

```
int x = 10;
int y = 10;

x = x * 2 + 2;
System.out.print("x: " + x);

x *= 2 + 2;
y += 7;

System.out.print("x: " + x);
System.out.println("y: " + y);
```

- a) x: 22, x: 88, y: 17 b) x: 22, x: 46, y: 17
c) x: 22, x: 22, y: 17 d) x: 40, x: 82, y: 17

2. Which of the following options is correct for the condition: *Either x must be less than 3 or y must be greater than or equal to 4?* (CLO-3, SO-b)

- a) `if ((x < 3) && (y > 4))`
b) `if ((x < 3) & y >= 4)`
c) `if ((x < 3) || (y >= 4))`
d) `if ((x > 3) || (y <= 4))`

When the statement `int num = 6.7;` is compiled in Java, it will result in: (CLO-4, SO-a)

- a) Compilation error c) Runtime error
b) *num* being 6.7 d) *num* being 6

Which of the following rules is true when naming variables? (CLO-4, SO-b)

- a) Variable names cannot start with numbers
b) Variable names cannot have symbols in them
c) Variable names cannot start with an underscore (`_`) symbol
d) Variable names can only contain letters and numbers

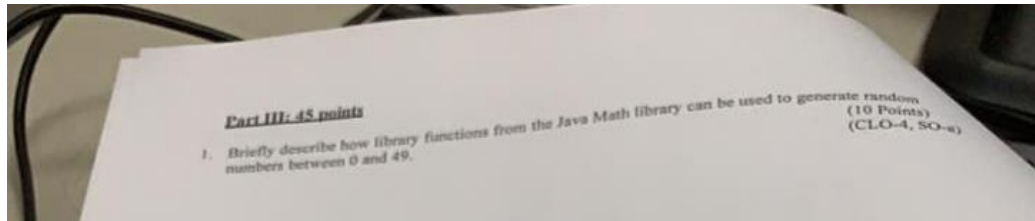
The statement below will generate a number between what range?

```
a = 20.0 + (Math.random() * 40);
```

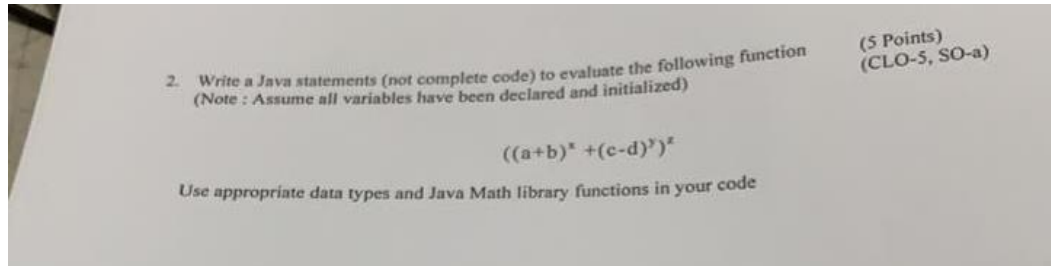
(CLO-4, SO-b)

- a) 20-59 c) 20-39
b) 0-59 d) 0-39

- 1- موبایلین
2- c
3- a
4- a
5- a



```
int a = (int)(Math.random()*50);
```



```
double j = Math.pow( (Math.pow((a+b),x) + Math.pow((c-d),y)),z);
```

3. Write a complete Java program that performs the following

(30 points)
(CLO-6, SO-a)

Your code should read prices of 3 items from the user. The prices entered will not include VAT. Your code should calculate the total and charge 5% VAT on the total amount. If the total price of the items is greater than or equal to SAR183.50 but less than SAR215.25, a 8% discount needs to be given on the total price. If the price is greater than or equal to SAR215.25, a 12% discount needs to be given. For items less than SAR183.50 no discount is provided. You code should display the final price (including VAT and discount, if applicable) to the user.

Sample Output

Please enter prices of three items : 100.25 20.5 20.5
Your total price (in SAR) is 148.3125

Please enter prices of three items : 100 40 60
Your total price (in SAR) is 193.2

Please enter prices of three items : 200 200 200
Your total price (in SAR) is 554.4

```
/* program also the command line arguments */
public static void main(String[] args) {
    double result;
    Scanner in = new Scanner(System.in);
    System.out.print("Please enter price of three items : ");
    double item1 = in.nextDouble();
    double item2 = in.nextDouble();
    double item3 = in.nextDouble();
    double total = item1 + item2 + item3;
    double total_VAT = total + total * (5.0 / 100.0);
    //System.out.println(total + " " + total_VAT);
    if (total_VAT >= 183.50 && total_VAT < 215.25) {
        result = total_VAT - total_VAT * (8.0 / 100.0);
    } else if (total_VAT >= 215.25) {
        result = total_VAT - total_VAT * (12.0 / 100.0);
    } else {
        result = total_VAT;
    }

    System.out.println("Your total price (in SAR) is " + result);
}
```

(10 Points)
(CLO-13, SO-a)

Code	Error
1 import java.util.scanner;	
2	
3 public Class Hello {	
4	
5 Public static void main(String[] args) {	
6 Double a = 2;	
7 int b = 5.6;	
8	
9 b = a + 1;	
10	
11 if (a=b);	
12 b = a - 1;	
13 else	
14 b = 10.0	
15	
16 system.out.print("X : " + a);	
17 System.out.print(`Y : ' + b);	
18 }	
19 }	

```

import java.util.Scanner;
double a = 2;
int b = 5;
b=(int) a+1;
if (a==b)
b=(int)a-1;
b=10;
System.out.print("X : "+a);
System.out.println("y : "+b);

```